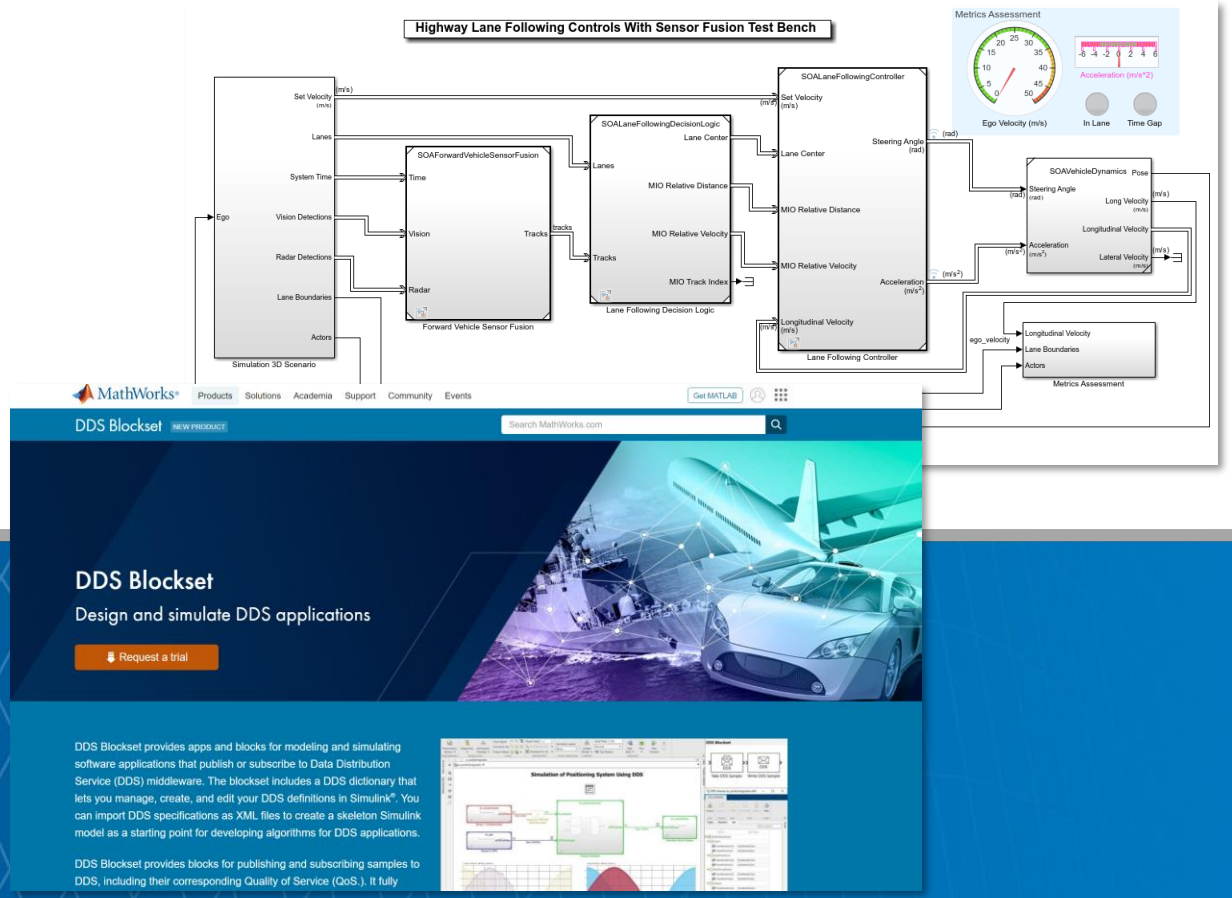
从Simulink模型生成C++代码，并通过消息接口集成到SOA框架

王鸿钧，资深应用工程师
hongjunw@mathworks.com
MathWorks 中国



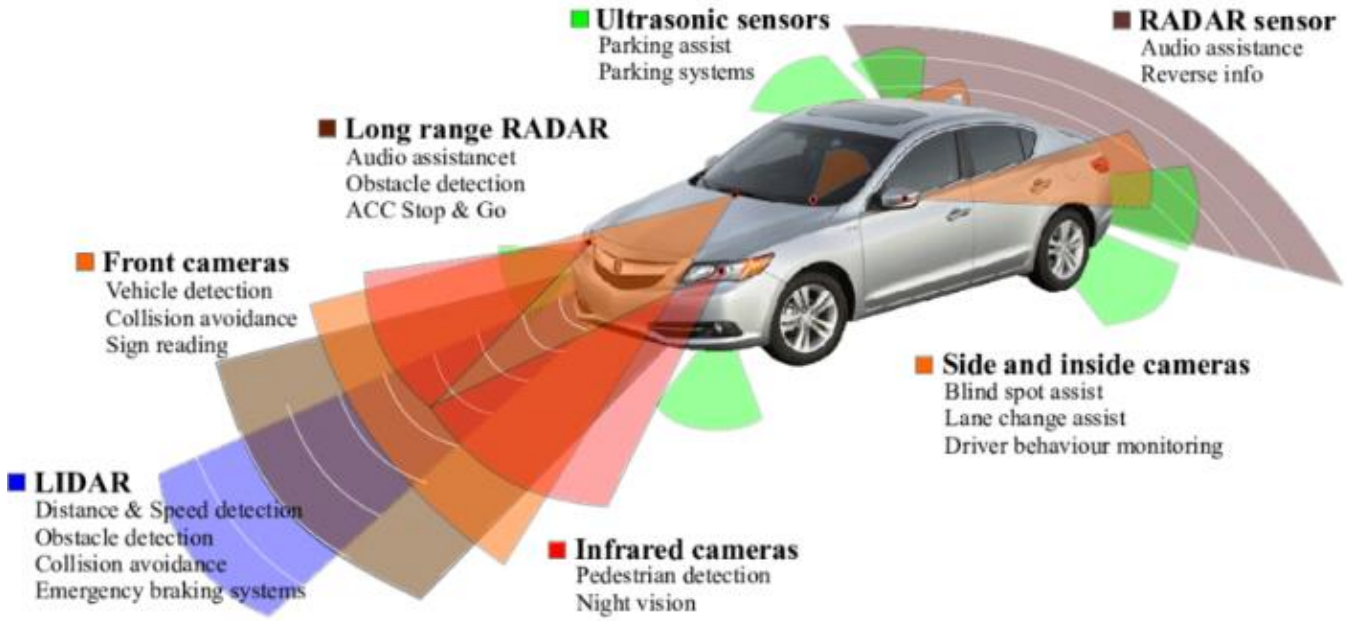
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自动驾驶对汽车电子系统和软件的挑战

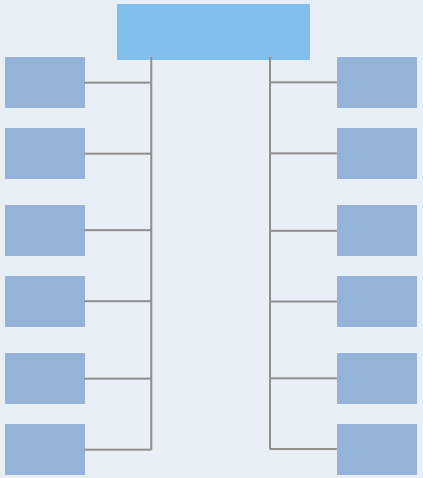
自动驾驶与网联给汽车带来巨大的附加价值

汽车电子系统和软件也面临大量的复杂挑战

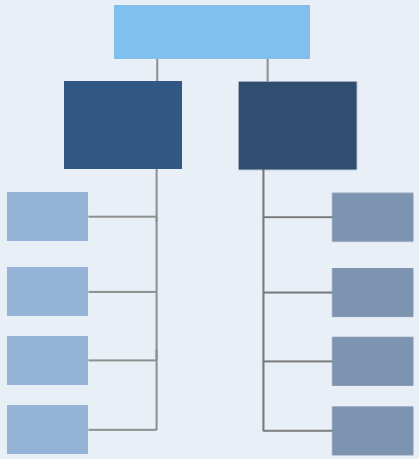




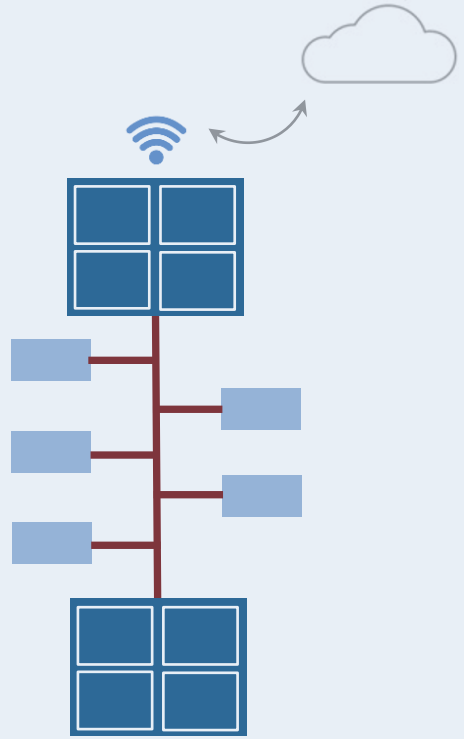
电子电气架构进化 – 算力向高性能车载计算机集中



- ~100 ECUs
- 1个功能对应1个 ECU
- 繁复的线束



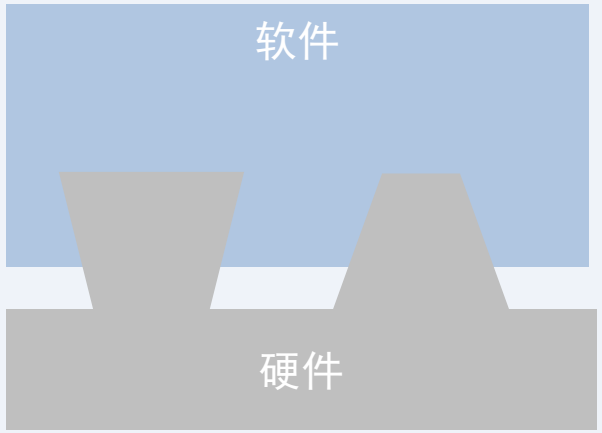
- 域控制器
- 多核 ECUs
- 网关和网络



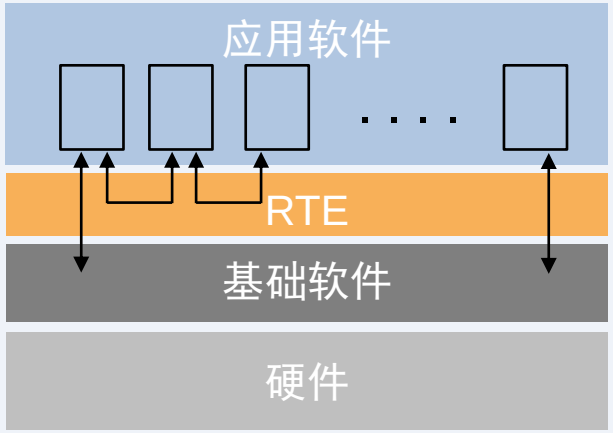
- 车辆/区域控制器
- 多核异构（CPU/GPU/FPGA）
- 高速以太网

分布 → 集中 → 高性能高连接

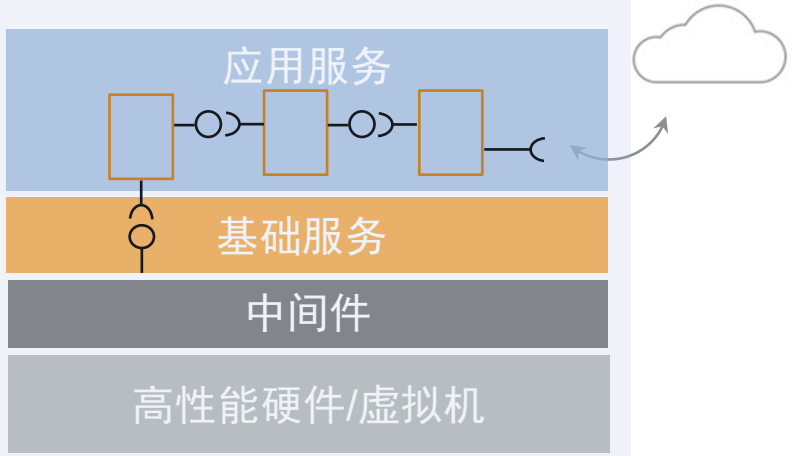
软件架构演变 – 面向服务的架构（SOA）



- 软硬件高度耦合
- 没有标准 APIs
- 软件复用程度低



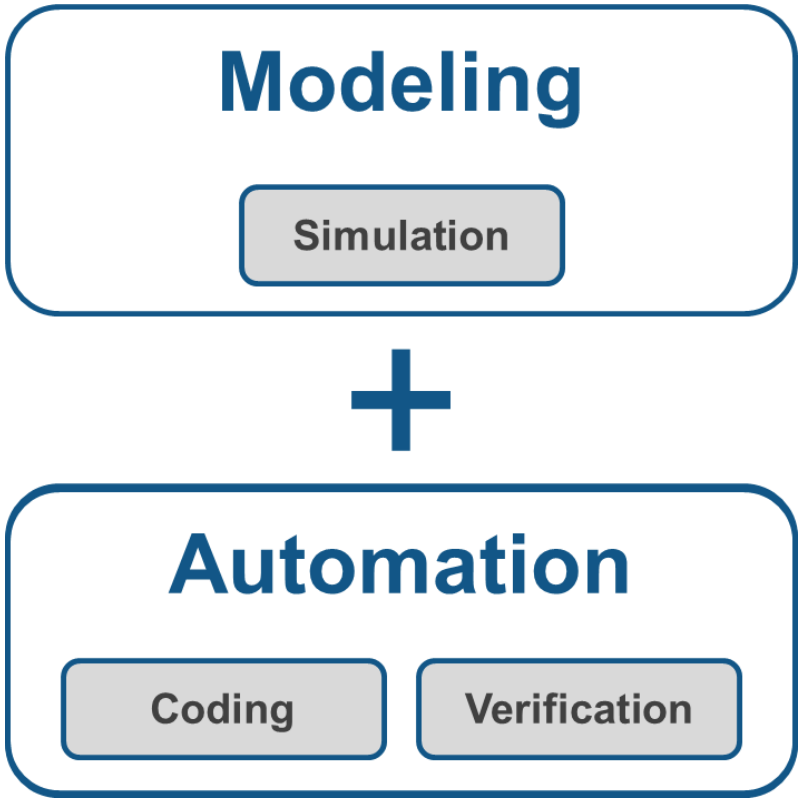
- 静态软件组件分配
- 基于信号通讯
- 硬件抽象
- 整体升级



- 动态服务发现
- 面向服务的通讯
- 更高级别的硬件抽象
- 选择性升级(OTA)

软硬耦合 → 基于组件 → 面向服务

解决方案 – 面向SOA的基于模型设计



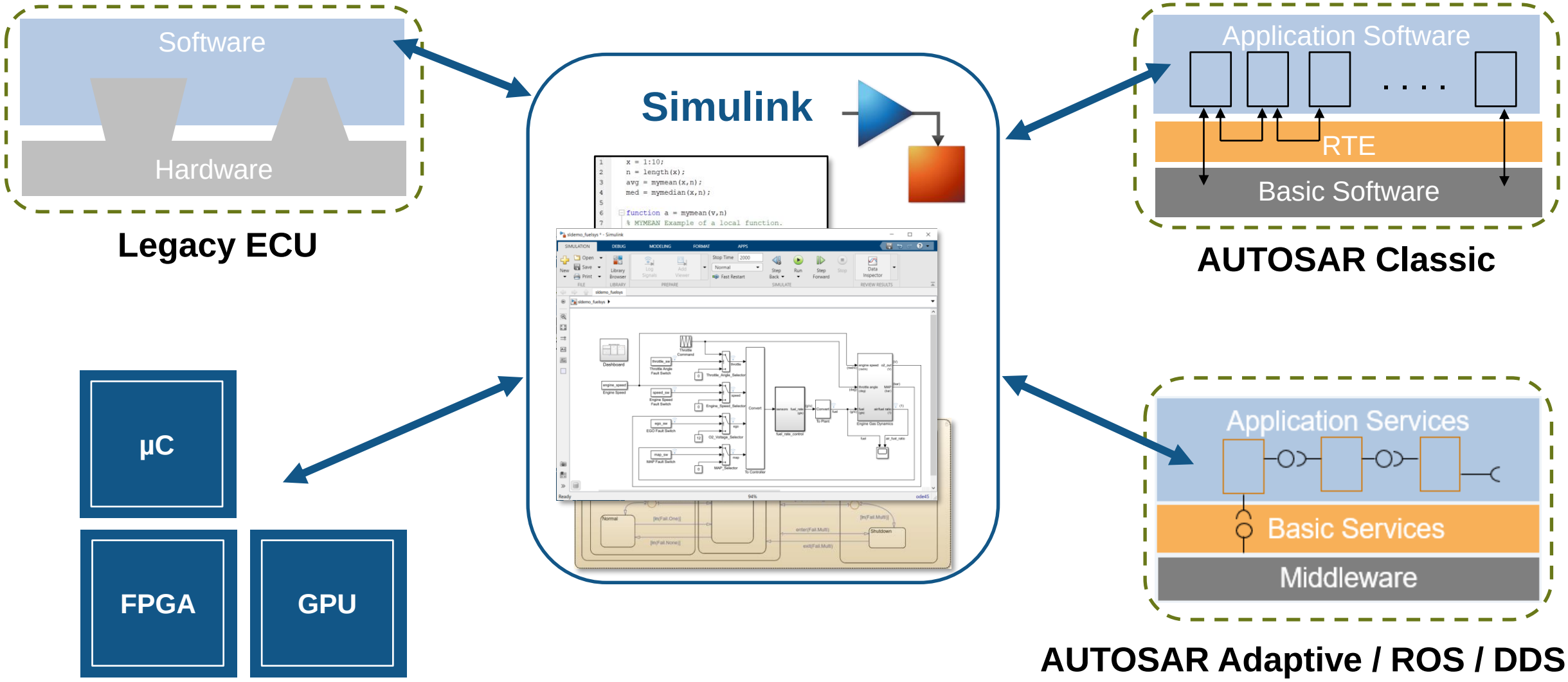
设计和仿真SOA应用

- 图形化的直观协同软件架构设计环境
- Simulink提供语义支持建模和仿真SOA应用

加强自动代码生成和验证支持

- 行业标准支持
- 异构目标硬件平台
- 集成人工智能算法

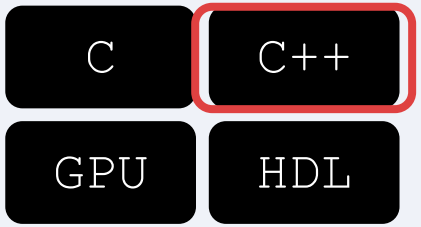
关键概念 – 以算法为中心


Software
Hardware
Legacy ECU
Simulink
Application Software
RTE
Basic Software
AUTOSAR Classic
µC
FPGA
GPU
Application Services
Basic Services
Middleware
AUTOSAR Adaptive / ROS / DDS

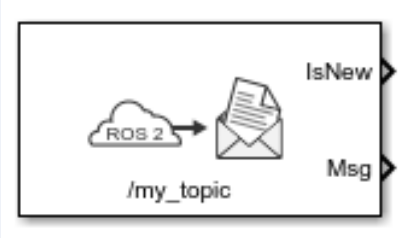
使用Simulink开发SOA软件的基础

MATLAB在自动驾驶软件工程的应用

代码生成



ROS / ROS 2.0



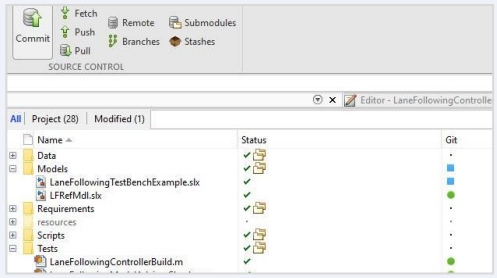
AUTOSAR



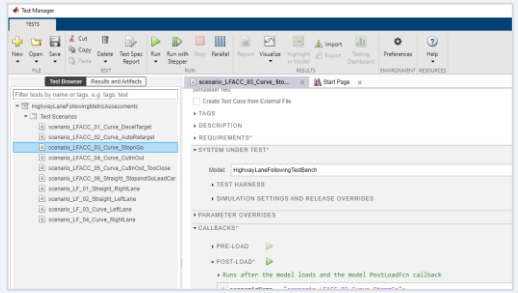
DDS



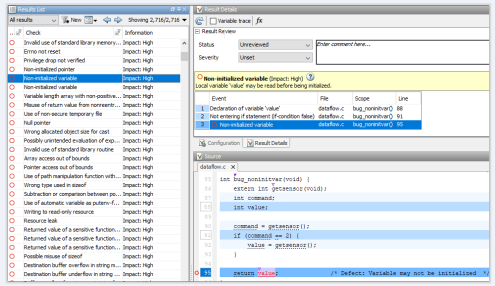
持续集成



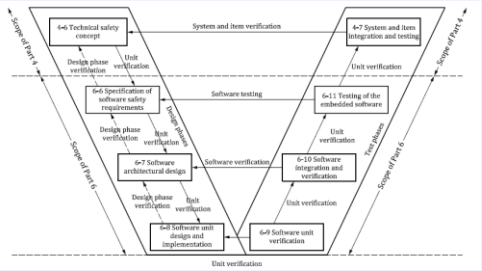
自动化测试



代码分析

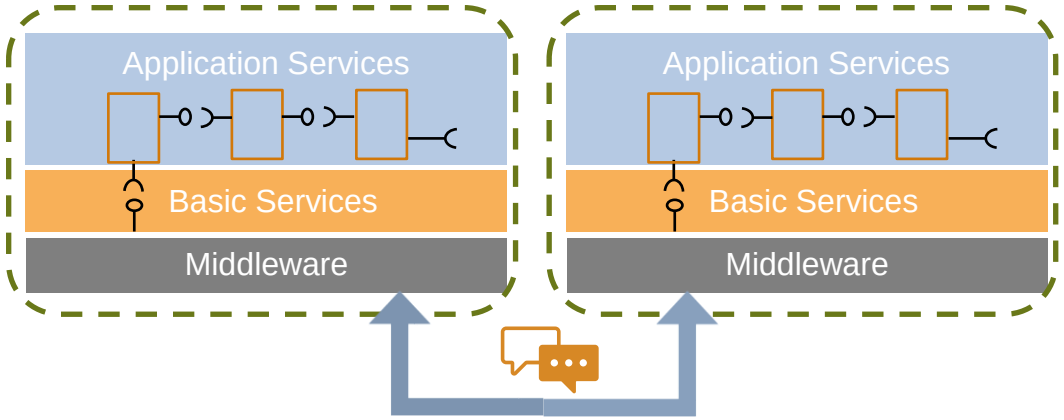


ISO 26262

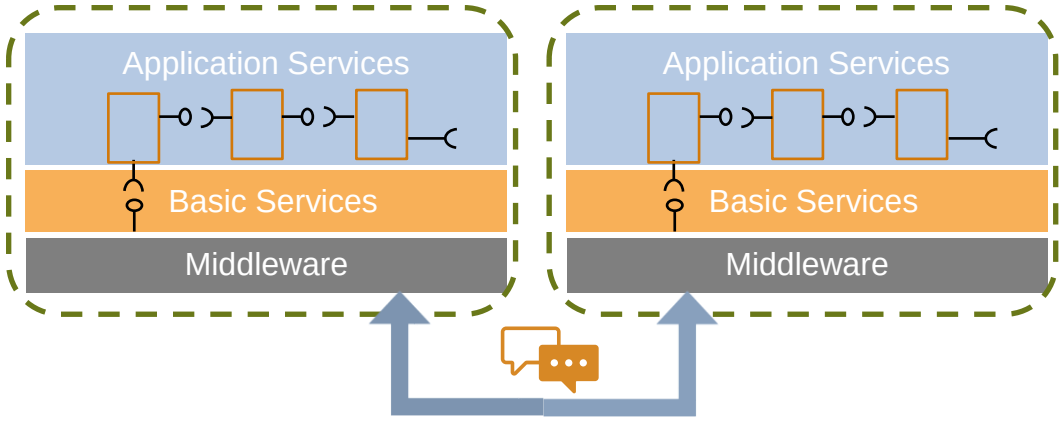


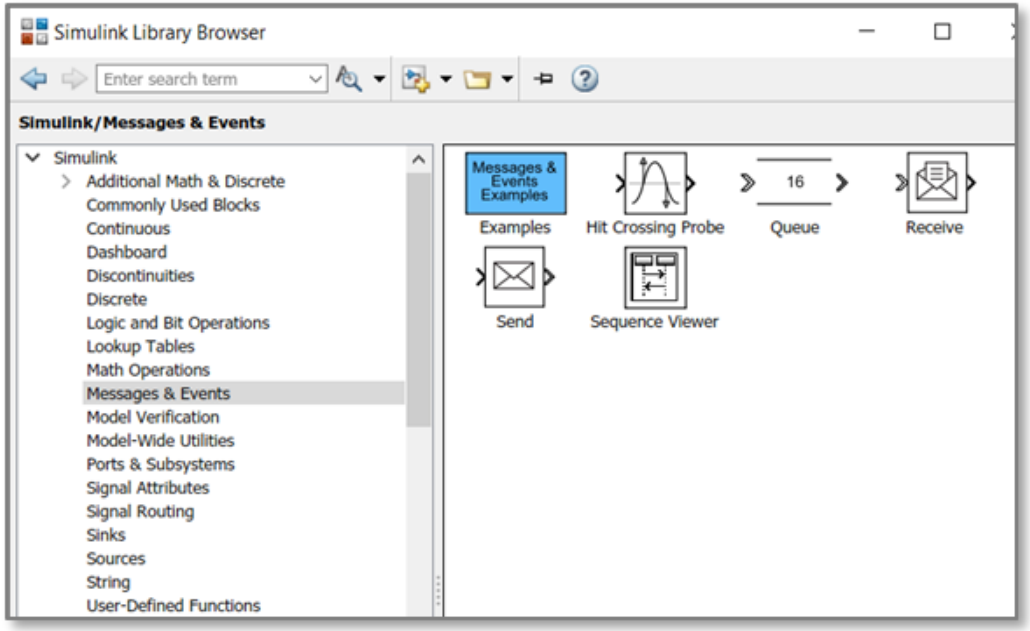
相关工具箱: MATLAB Coder, Embedded Coder, GPU Coder, HDL Coder,
ROS Toolbox, AUTOSAR Blockset, DDS Blockset,
Simulink Test, Simulink Coverage, Polyspace, IEC Certification Kit,

面向服务的通讯建模支持 – Simulink Messages

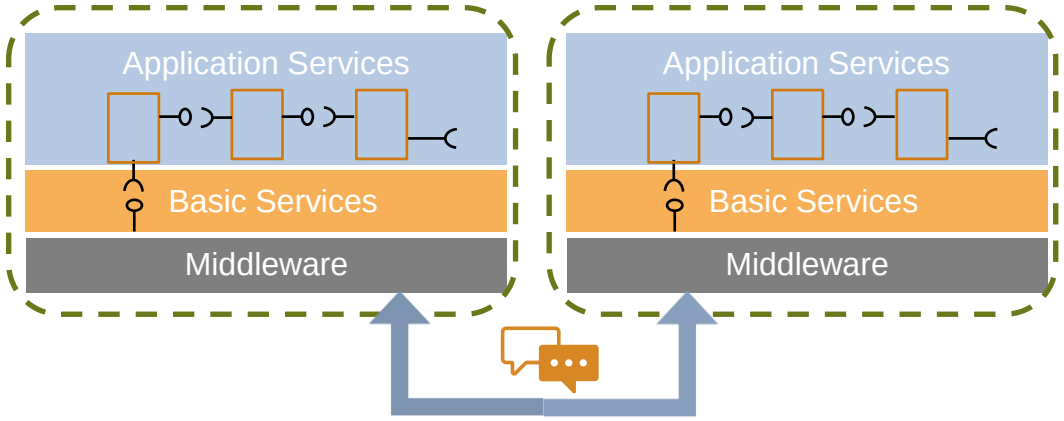


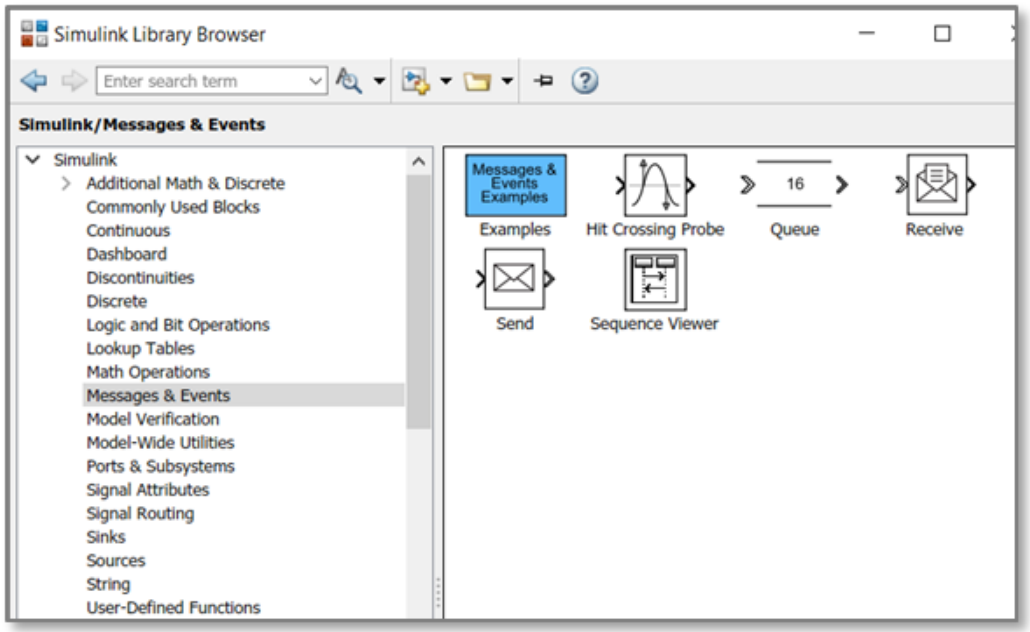
面向服务的通讯建模支持 – Simulink Messages

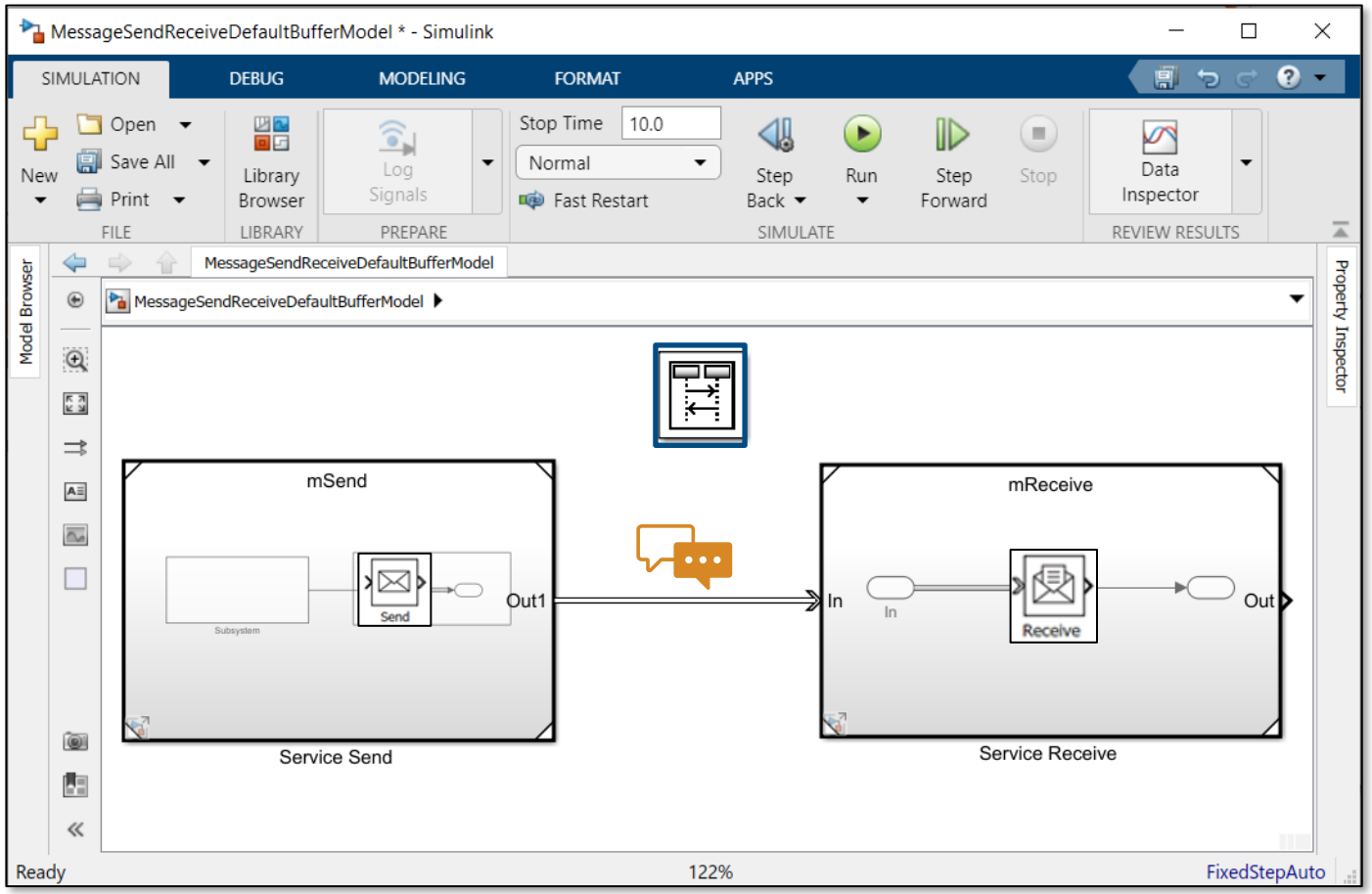




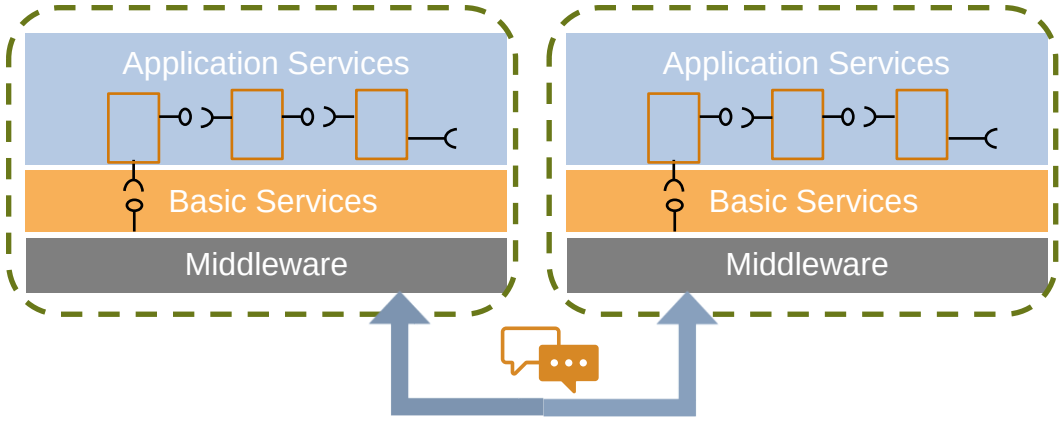
面向服务的通讯建模支持 – Simulink Messages

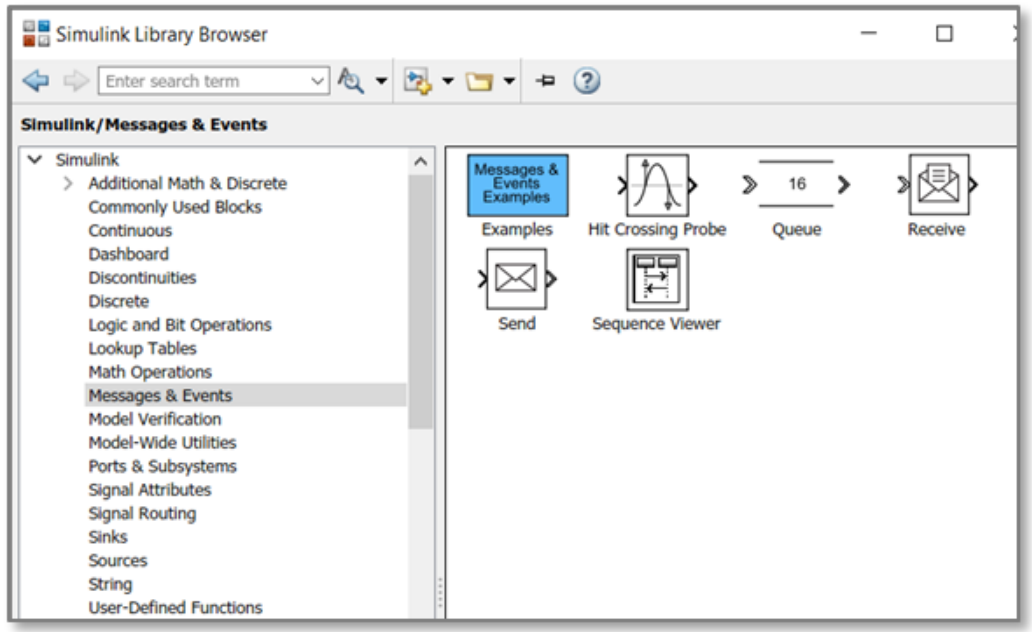


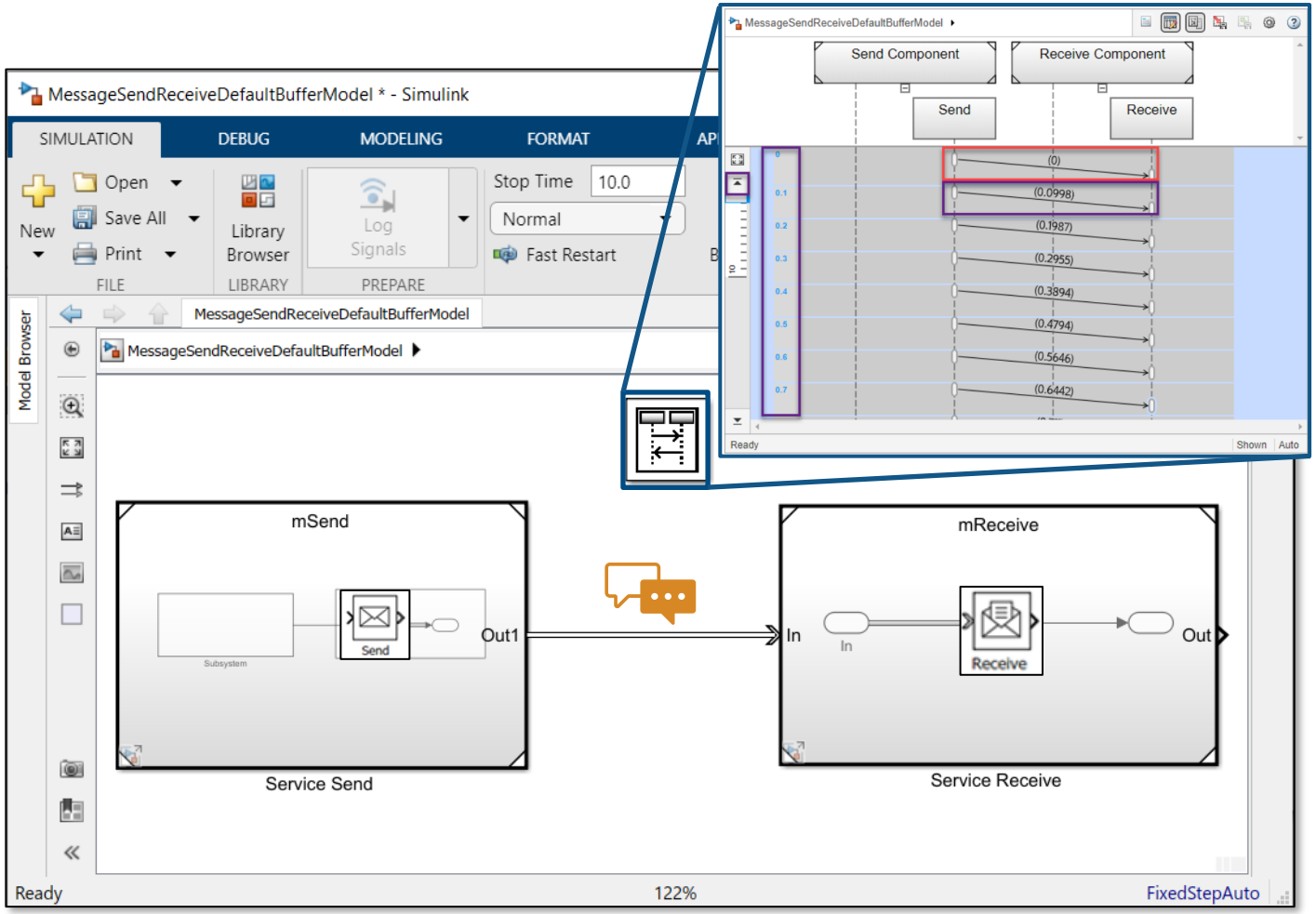




面向服务的通讯建模支持 – Simulink Messages



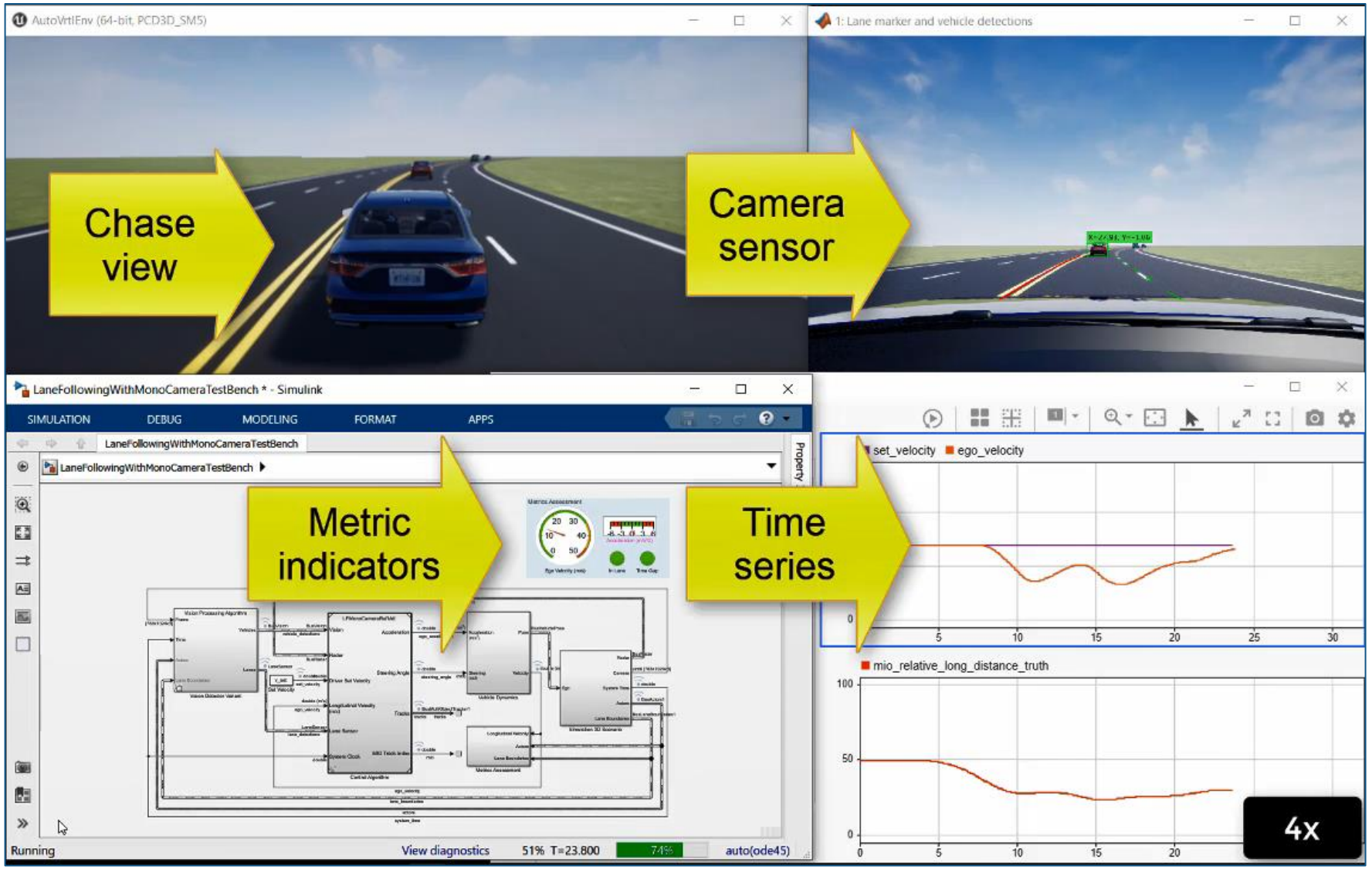




使用消息发送和接收模块建模面向服务的通讯

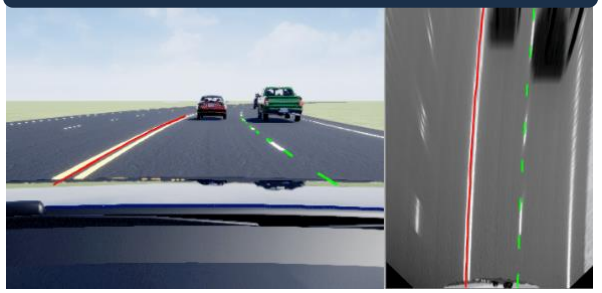
高速公路车道跟随系统的基于消息建模

高速公路车道跟随 – Highway Lane Following



高速公路车道跟随 – Highway Lane Following

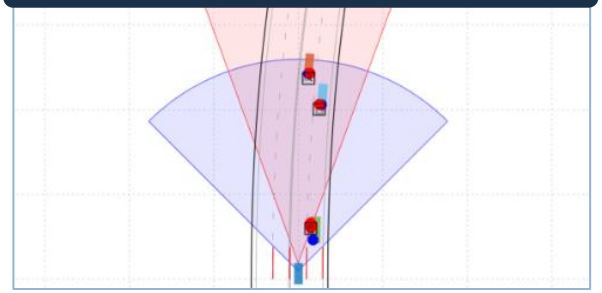
车道线检测



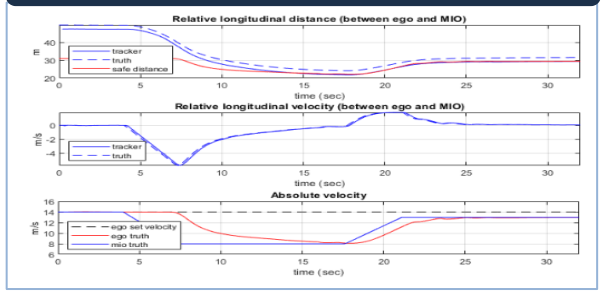
车辆检测

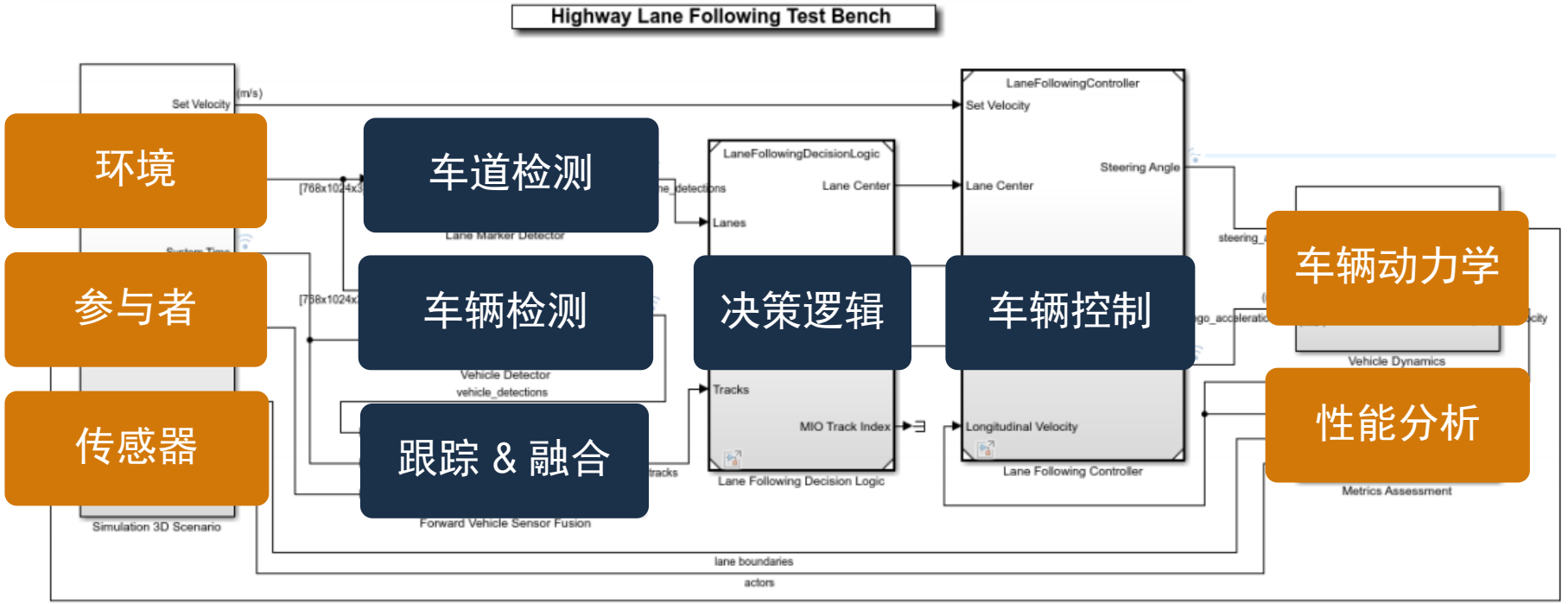


跟踪 & 融合



决策 & 控制

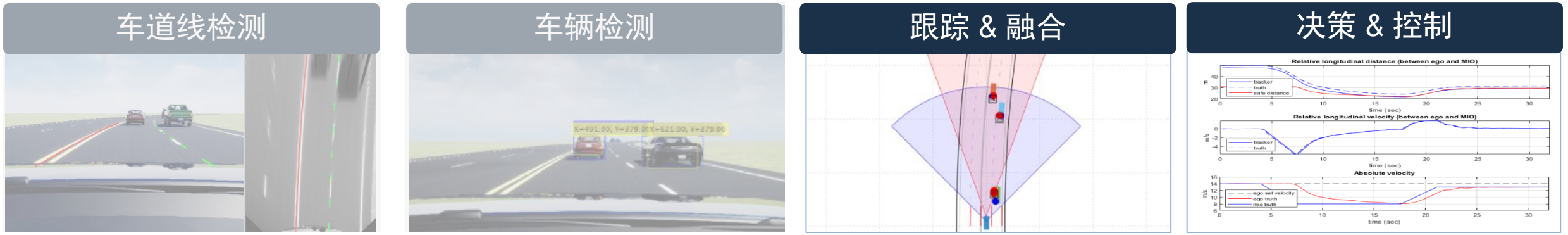




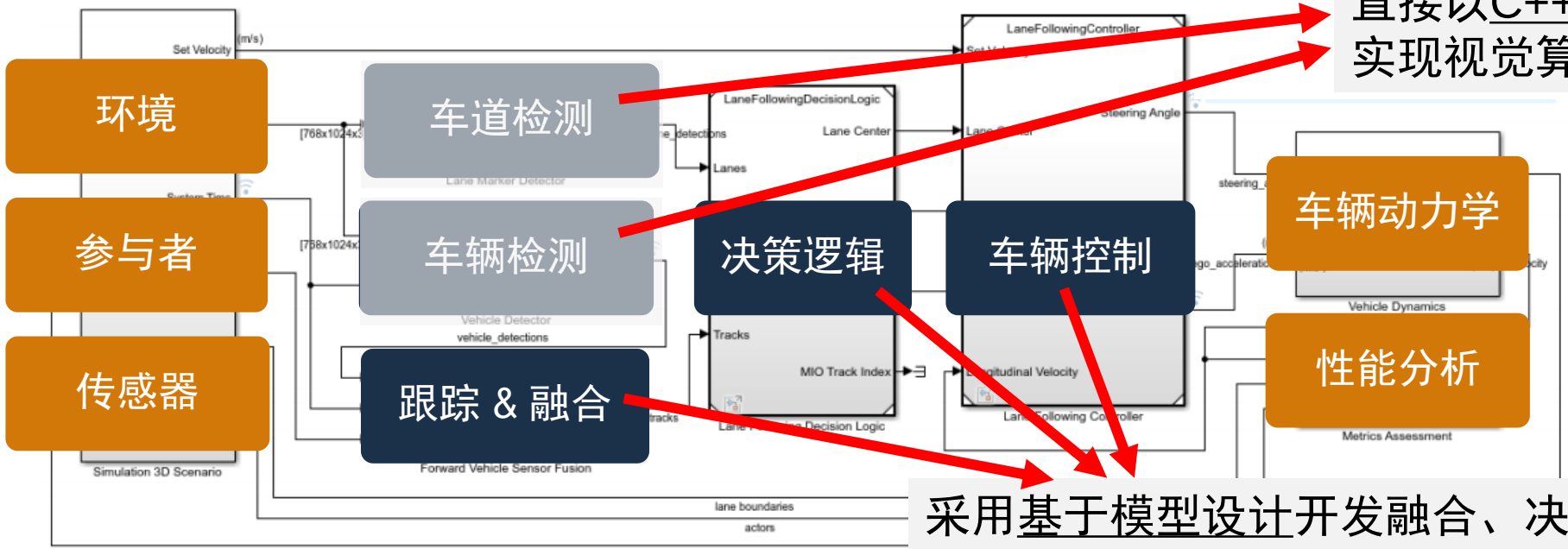
[Highway Lane Following]

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高速公路车道跟随 – Highway Lane Following



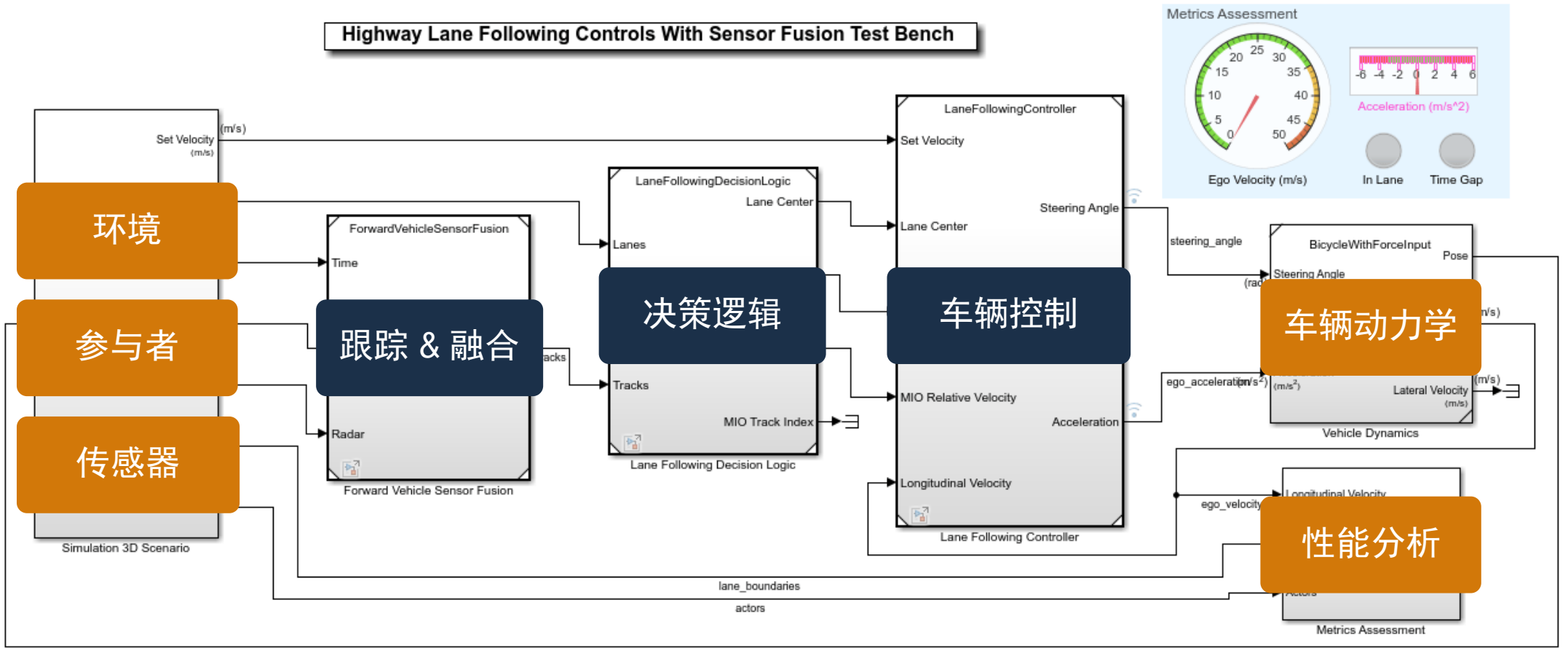
不同软件团队、供应商可以选用不同的开发方式



[Highway Lane Following]

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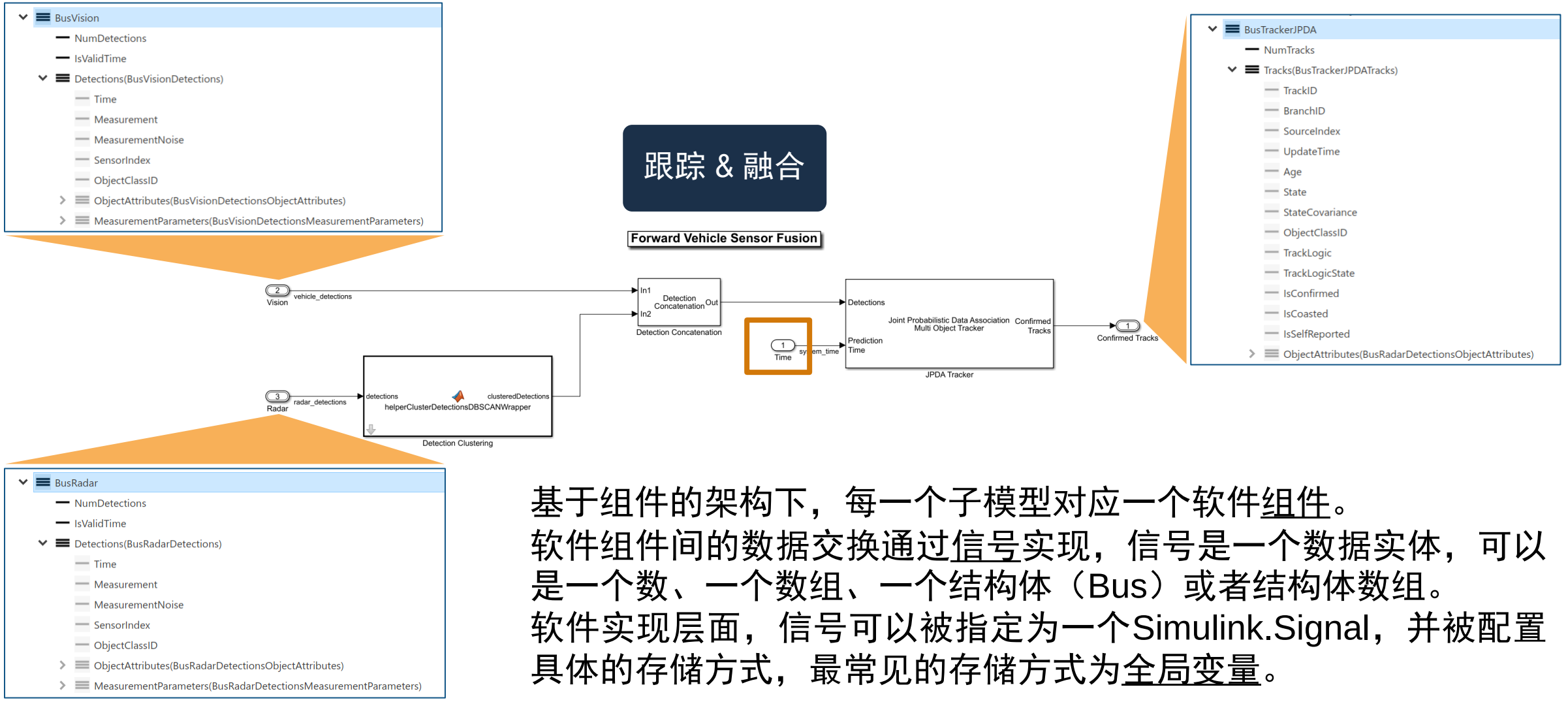
高速公路车道跟随 – 对融合、决策、控制组件建模和在环测试



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[Automate Testing for Highway Lane Following Controls and Sensor Fusion]

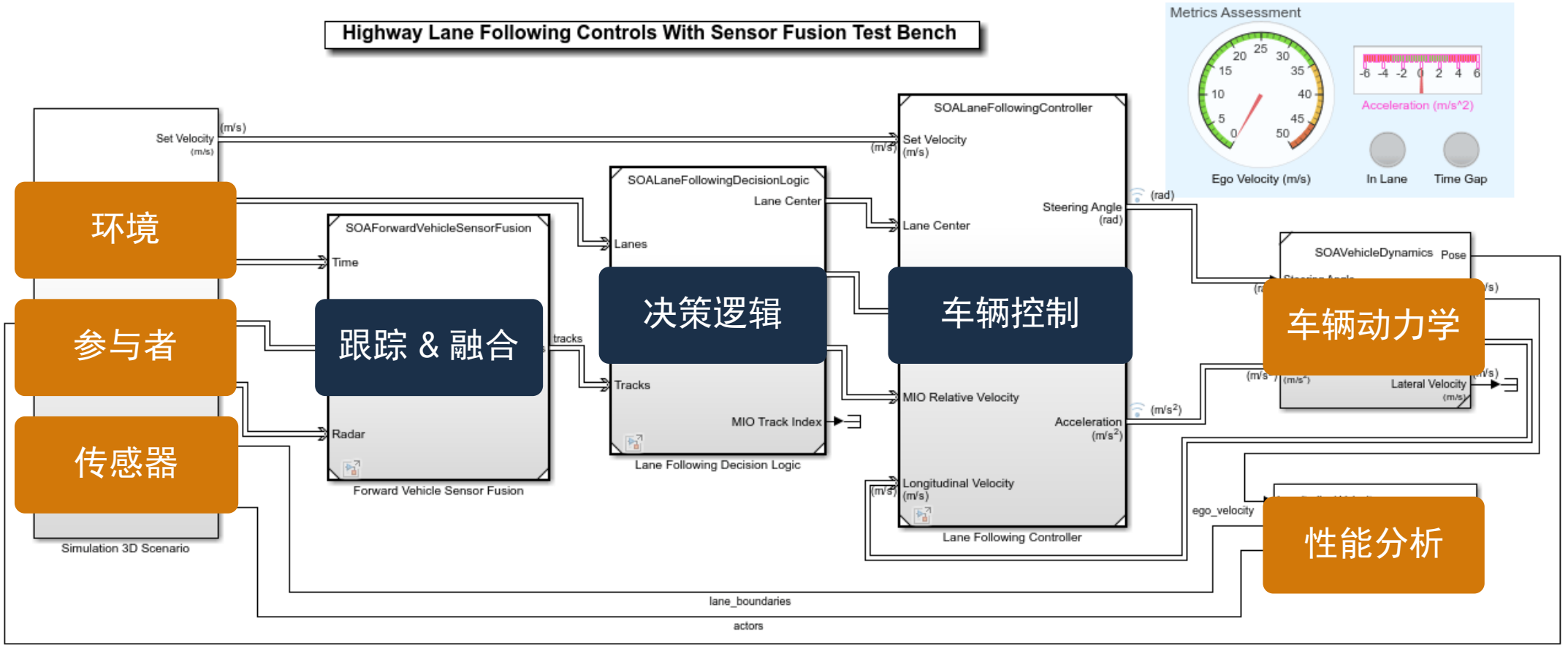
高速公路车道跟随 – 软件组件间的信号接口



基于组件的架构下，每一个子模型对应一个软件组件。
软件组件间的数据交换通过信号实现，信号是一个数据实体，可以是一个数、一个数组、一个结构体（Bus）或者结构体数组。
软件实现层面，信号可以被指定为一个Simulink.Signal，并被配置具体的存储方式，最常见的存储方式为全局变量。

[Automate Testing for Highway Lane Following Controls and Sensor Fusion]

高速公路车道跟随 – 基于消息的建模

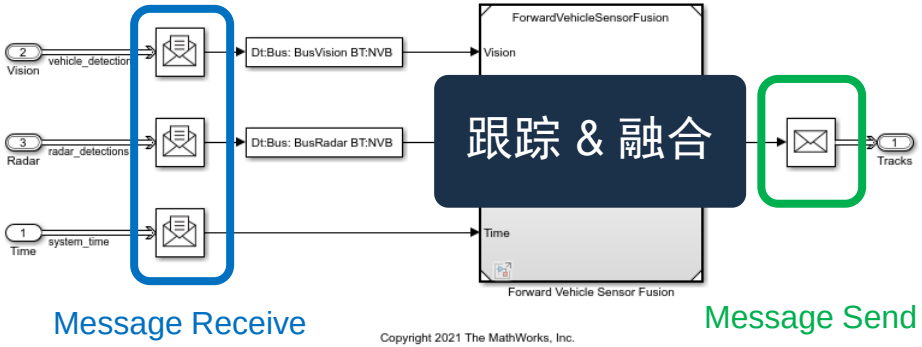


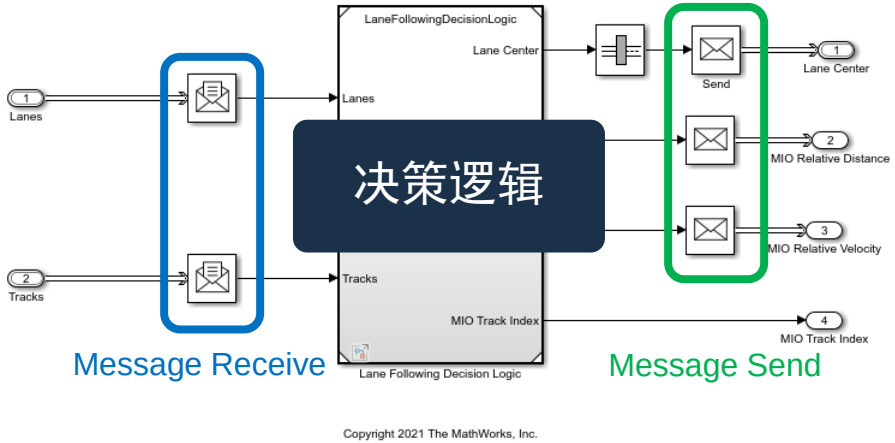
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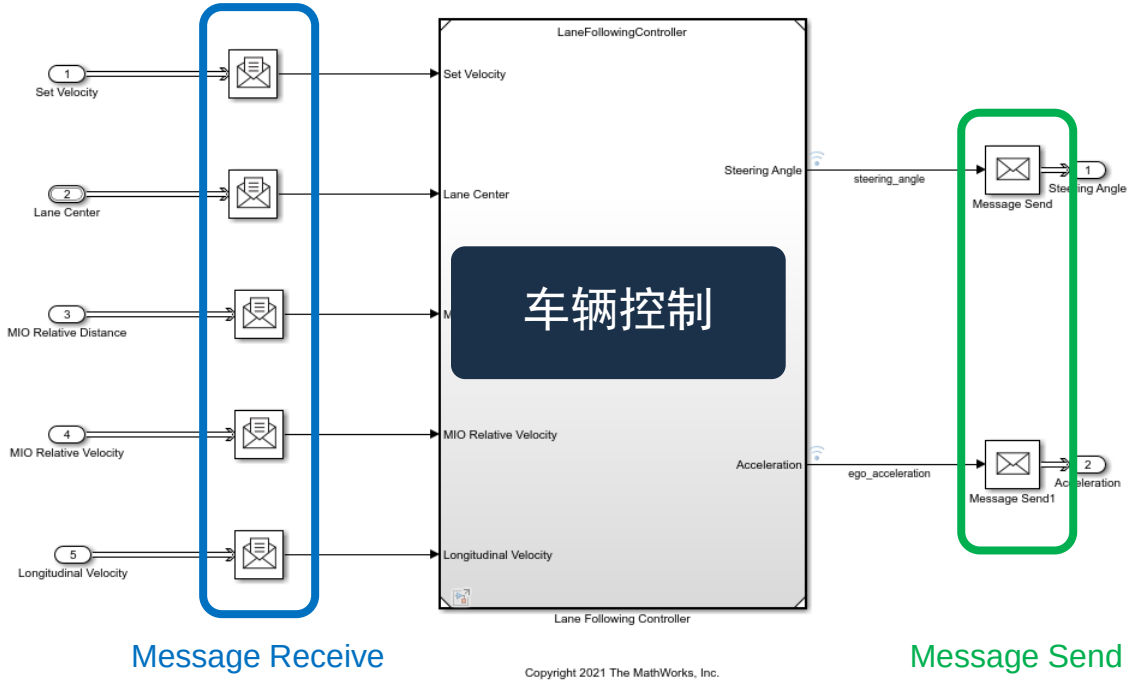
[Generate C++ Message Interfaces for Lane Following Controls and Sensor Fusion]

R2021b

高速公路车道跟随 – 软件应用间的消息接口







面向服务的架构下，每一个子模型对应一个软件应用。
为了建立软件应用的消息接口，在输入和输出端口分别添加消息的Receive和Send模块。
软件实现层面，应用间的数据交换通过操作系统或中间件的通讯服务完成，提供Receive和Send的具体方法。

[Generate C++ Message Interfaces for Lane Following Controls and Sensor Fusion]

R2021b

生成C++代码并运用于SOA中间件

高速公路车道跟随 – 从传感器融合子模型生成C++代码

设置代码生成选项，通过Embedded Coder生成C++代码

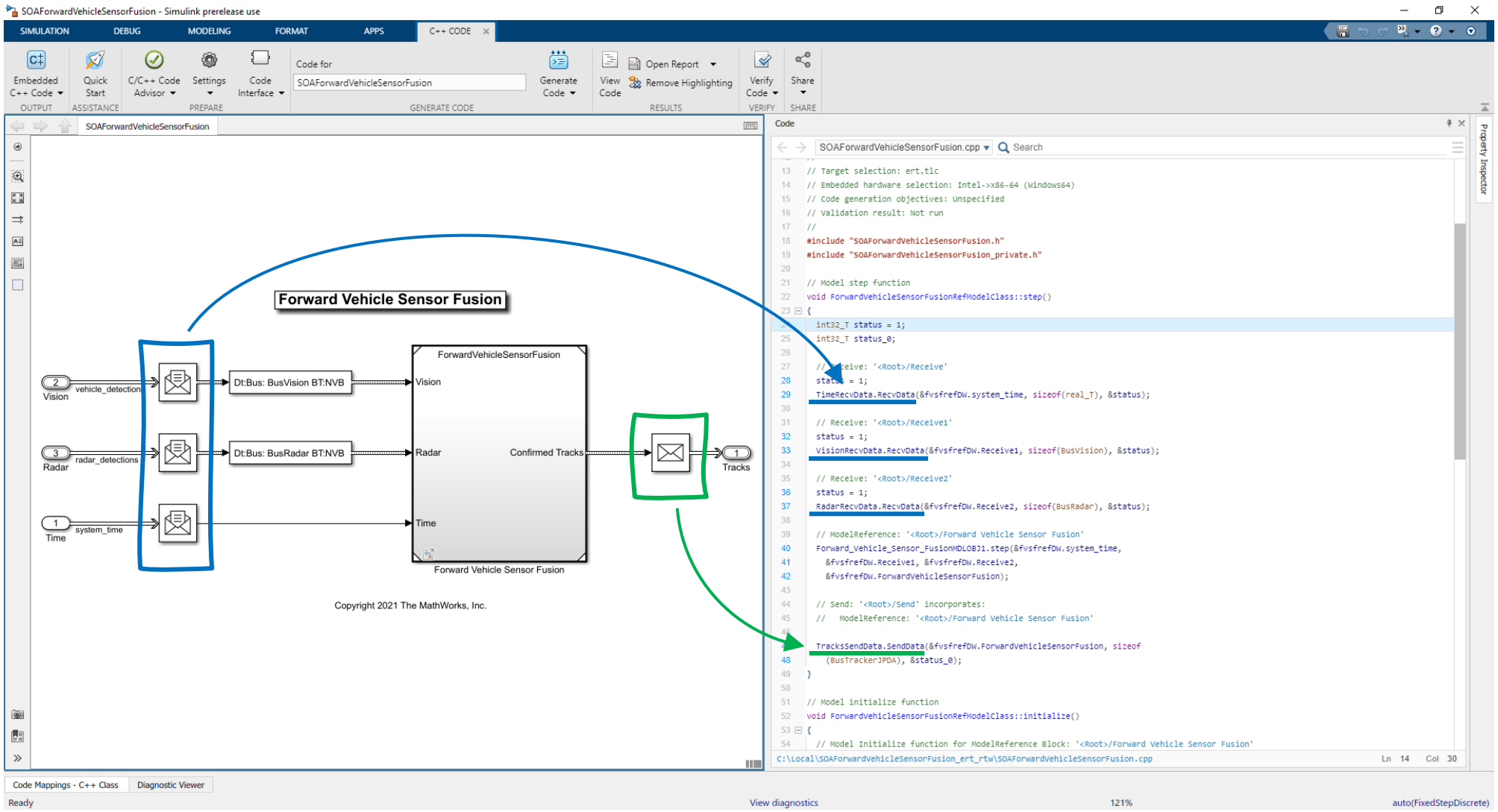
Model configuration parameters:

Parameter	Value	Description
'SystemTargetFile'	'ert.tlc'	'Code Generation>System target file'
'TargetLang'	'C++'	'Code Generation>Language'
'SolverType'	'Fixed-step'	'Solver>Type'
'FixedStep'	'auto'	'Solver>Fixed-step size (fundamental sample time)'
'EnableMultiTasking'	'on'	'Solver>Treat each discrete rate as a separate task'
'ProdLongLongMode'	'on'	'Hardware Implementation>Support long long'
'BlockReduction'	'on'	'Simulation Target>Block reduction'
'MATLABDynamicMemAlloc'	'on'	'Simulation Target>Simulation Target>Dynamic memory allocation in MATLAB functions'
'OptimizeBlockIOStorage'	'on'	'Simulation Target>Signal storage reuse'
'InlineInvariantSignals'	'on'	'Simulation Target>Inline invariant signals'
'BuildConfiguration'	'Faster Runs'	'Code Generation>Build configuration'
'RTWVerbose'	'off'	'Code Generation>Verbose build'
'CombineSignalStateStructs'	'on'	'Code Generation>Interface>Combine signal/state structures'
'SupportVariableSizeSignals'	'on'	'Code Generation>Interface>Support variable-size signals'
'CodeInterfacePackaging'	'C++ class'	'Code Generation>Interface>Code interface packaging'
'GenerateExternalIOAccessMethods'	'Method'	'Code Generation>Interface>Data Member Visibility>External I/O access'
'EfficientFloat2IntCast'	'on'	'Code Generation>Optimization>Remove code from floating-point to integer conversions that wraps out-of-range values'
'ZeroExternalMemoryAtStartup'	'off'	'Code Generation>Optimization>Remove root level I/O zero initialization (inverse logic)'
'CustomSymbolStrGlobalVar'	'NM'	'Code Generation>Symbols>Global variables'
'CustomSymbolStrType'	'NM_T'	'Code Generation>Symbols>Global types'
'CustomSymbolStrField'	'NM'	'Code Generation>Symbols>Field name of global types'
'CustomSymbolStrFcn'	'APV_NM$F'	'Code Generation>Symbols>Subsystem methods'
'CustomSymbolStrTmpVar'	'NM'	'Code Generation>Symbols>Local temporary variables'
'CustomSymbolStrMacro'	'NM'	'Code Generation>Symbols>Constant macros'

[Generate C++ Message Interfaces for Lane Following Controls and Sensor Fusion]

R2021b

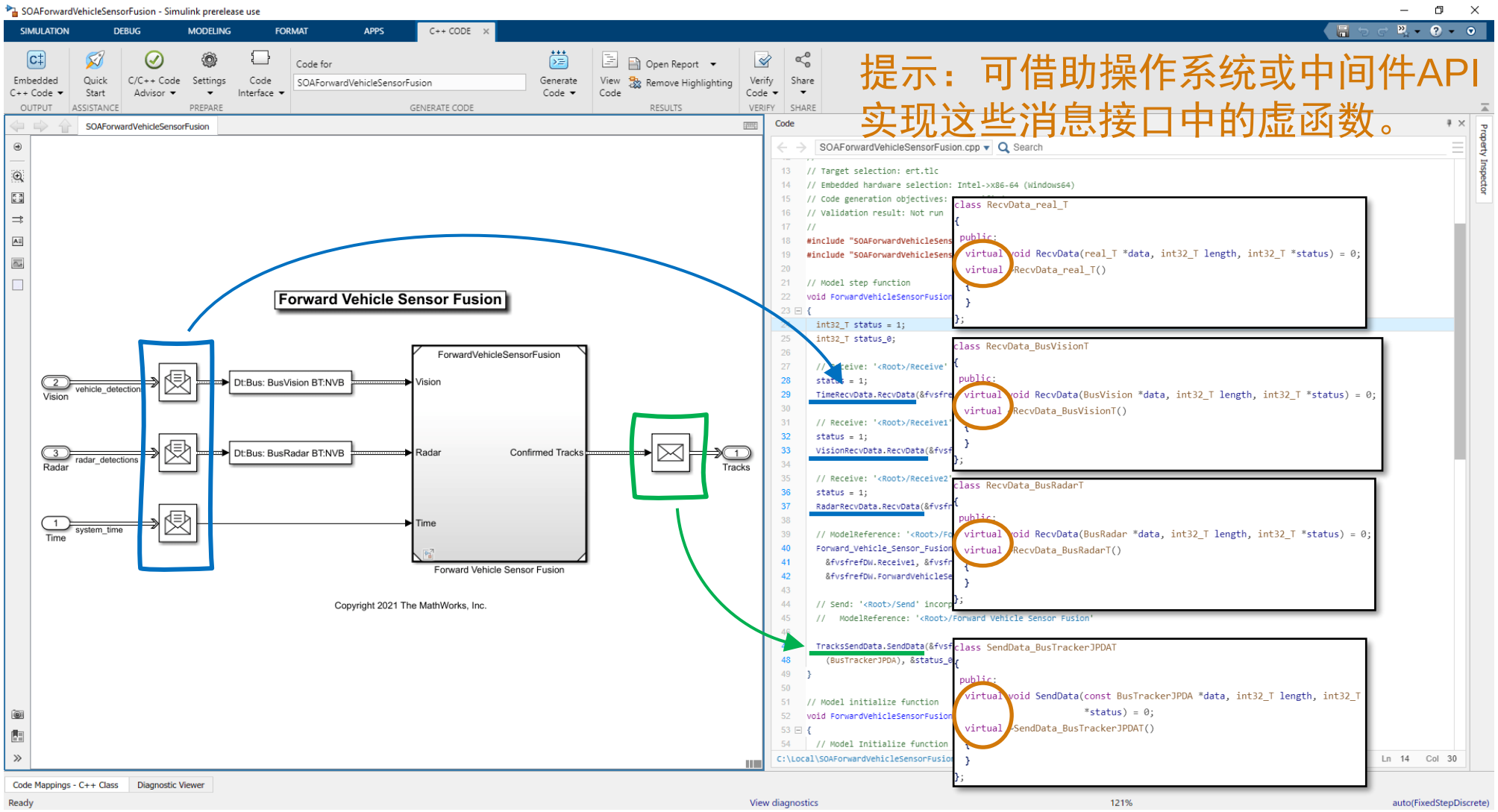
高速公路车道跟随 – 从传感器融合子模型生成C++代码



[Generate C++ Message Interfaces for Lane Following Controls and Sensor Fusion]

R2021b

高速公路车道跟随 – 从传感器融合子模型生成C++代码



提示：可借助操作系统或中间件API实现这些消息接口中的虚函数。

[Generate C++ Message Interfaces for Lane Following Controls and Sensor Fusion]

R2021b

手动集成到SOA示例 – 通过POSIX消息队列API收发消息

- Use Handwritten Code to Integrate C++ Messages with POSIX
- 建立Send Class实现消息发送

这是从Simulink生成C++代码并集成到任何SOA框架的一般方法！

手写代码实现消息发送（通过mq_send）

```
class accelSendModelClassSendData_real_T : public SendData_real_T{
    mqd_t msgQueue;

    public:
    // Use POSIX to send messages
    void SendData(const real_T* data, int32_T length, int32_T* status)
    {
        unsigned int priority = 1;
        *status = mq_send(msgQueue, (char*)data, length, priority);
    }

    // Set method for class POSIX queue
    void SetMQ(mqd_t mq)
    {
        msgQueue = mq;
    }
};
```

- 创建Send Object并关联到模型Object

实例化消息发送类，并关联到模型对象

```
// Create send object
static accelSendModelClassSendData_real_T Out1SendData_arg;

// Attach send object to sensor model to send messages
static accelSendModelClass rtObj( Out1SendData_arg);
```

R2021b

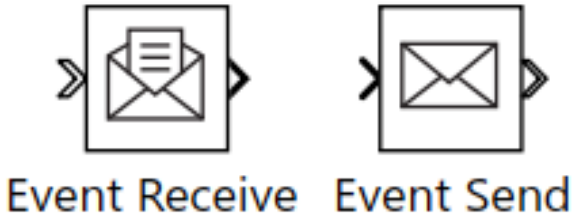
使用相关工具箱，自动化SOA集成过程

- 使用特定的Simulink库，进行消息接口建模
- 直接在生成的C++代码中自动调用对应中间件的通讯服务，实现消息的收发
- 完善的消息接口管理工具（类似于数据字典）

NEW!

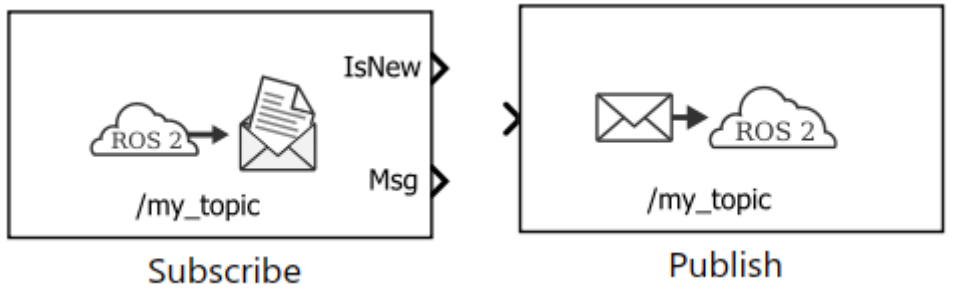
AUTOSAR Blockset

Design and simulate AUTOSAR software



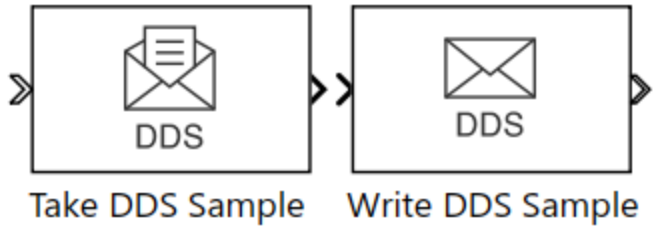
ROS Toolbox

Design, simulate, and deploy ROS-based applications



DDS Blockset

Design and simulate DDS applications

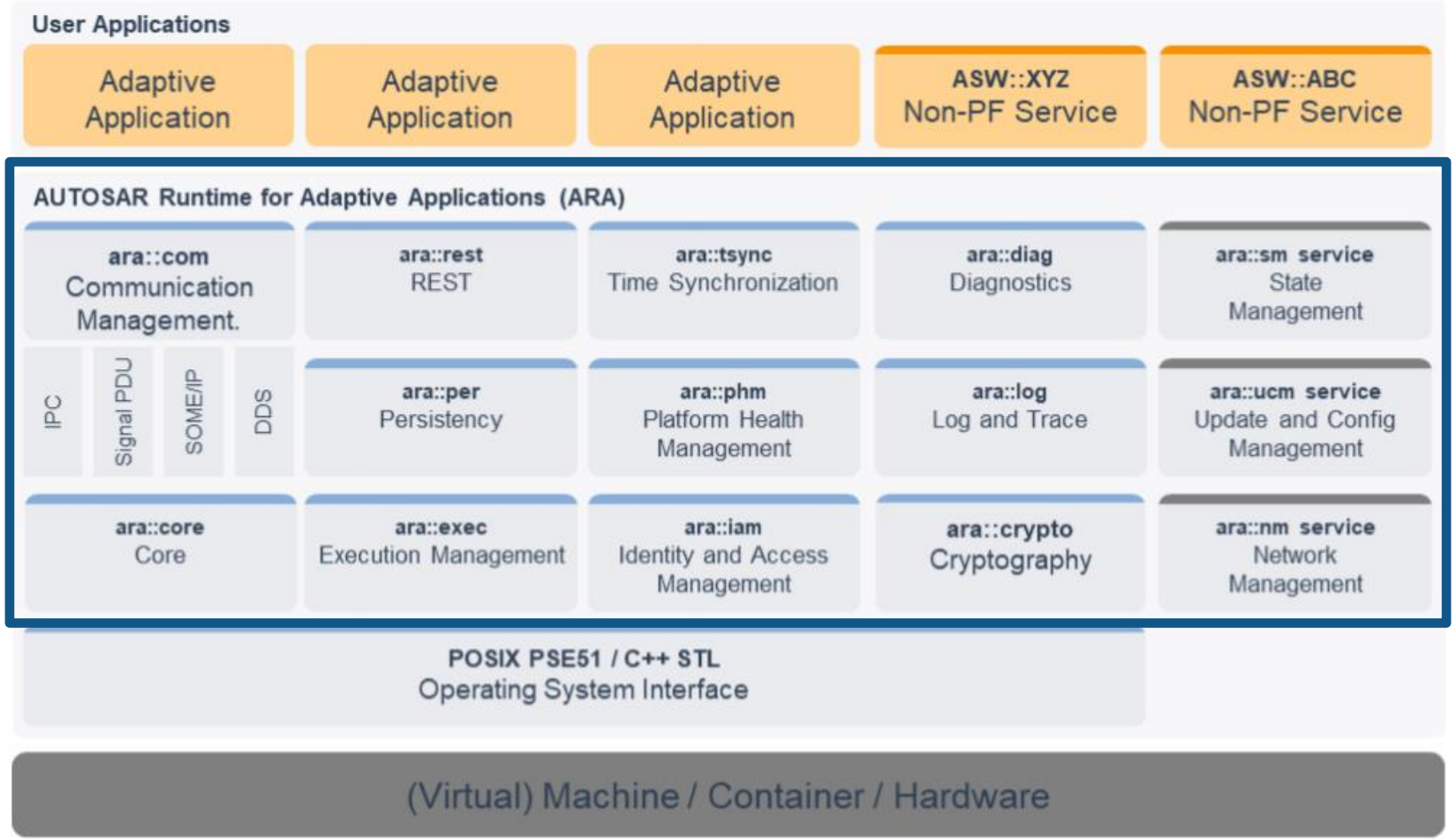


选择一种SOA解决方案 – 选择一种已有的生态系统作为平台

例如：

AUTOSAR Adaptive

- 服务和接口(ARA)
- 客户端（应用）和服务器的运行时动态链接
- 通讯中间件：
  SOME/IP 或 DDS



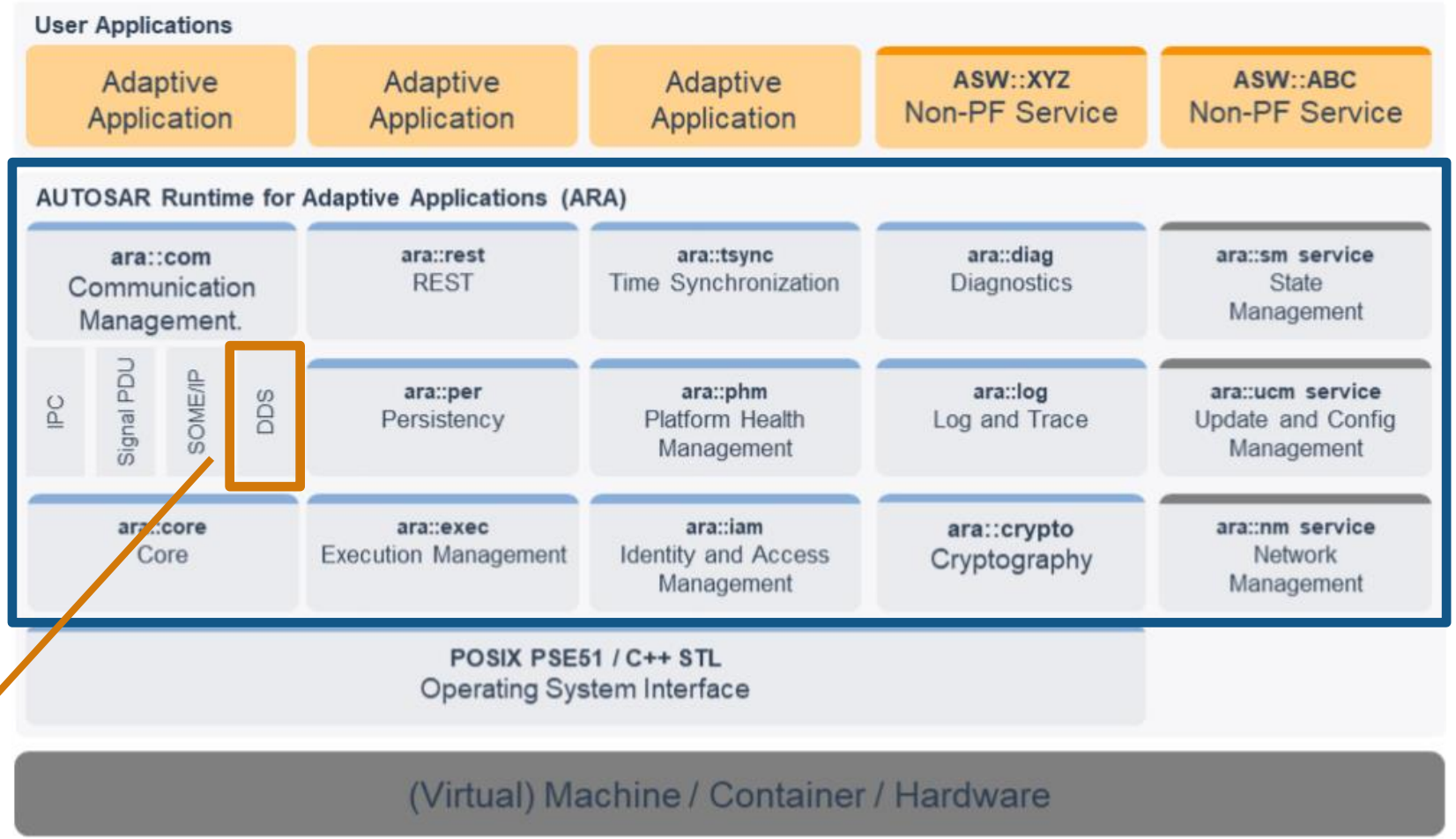
Source: www.autosar.org

选择一种SOA解决方案 – 也可基于标准的关键技术进行集成

AUTOSAR Adaptive

- 服务和接口(ARA)
- 客户端（应用）和服务器的运行时动态链接
- 通讯中间件：
  SOME/IP 或 DDS

不拘泥于某个特定平台，灵活开发自主的体系结构，包括使用分层数据总线



Source: www.autosar.org

全新DDS工具箱及其对RTI Connext的支持

数据分发服务（Data Distribution Service）

Data Distribution Service (DDS) 是一种面向实时系统，支持发布/订阅等通信方式的通信架构

DDS已被广泛应用于依赖实时数据交换的领域（例如自动驾驶、航空航天、机器人）

R2021a

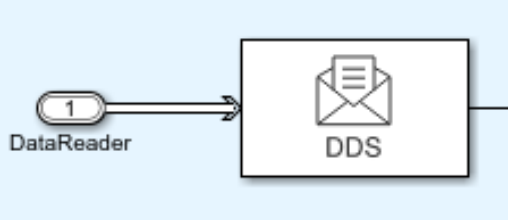


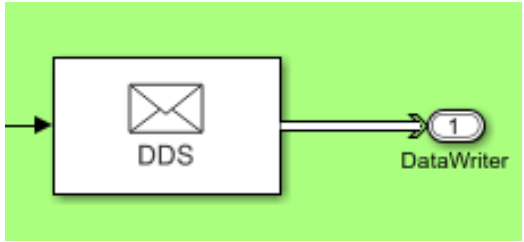
DDS Blockset 概要

R2021a

DDS应用程序建模和仿真

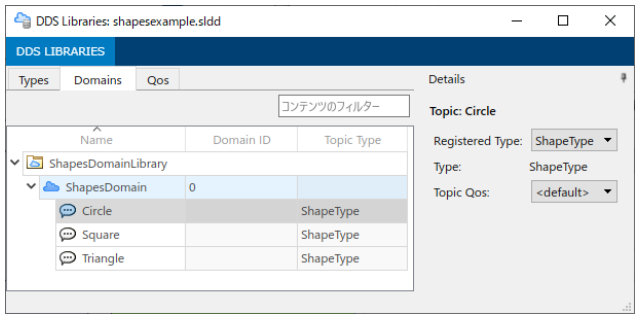
向DDS中间件Publish/Subscribe

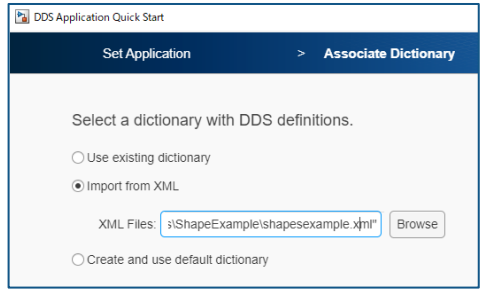




管理DDS相关的定义

可通过数据字典管理DDS相关信息，
并支持导入DDS相关定义





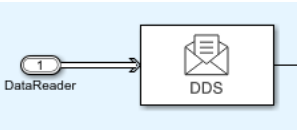
联合DDS商业工具共同开发

RTI官方提供了一个MATLAB插件，含评估版RTI Connext

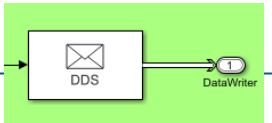
支持自动代码生成

C++/XML的自动生成，并支持编译为可执行文件

```
// Receive: '<Root>/Take DDS Sample'
rtb_x = 1;
DataReaderRecvData.RecvData(&shapesdemo_B.TakeDDSSample,
```

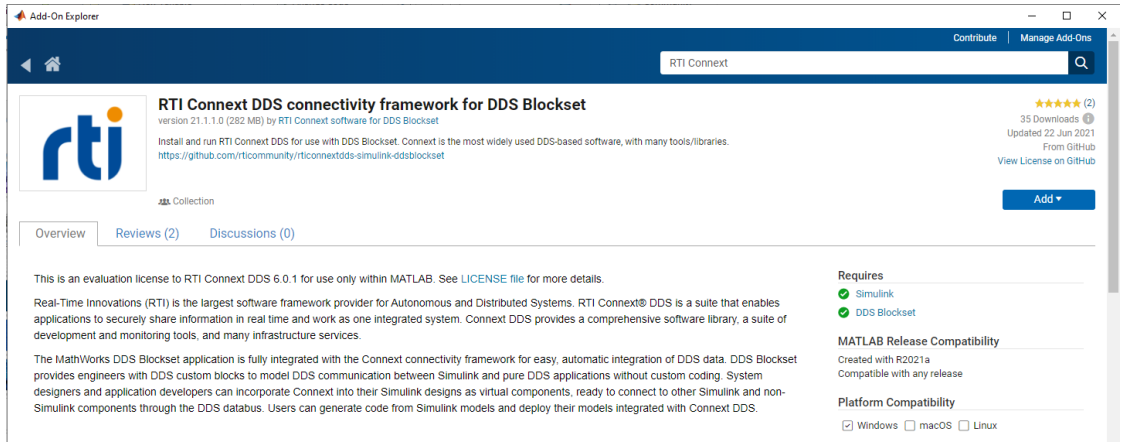


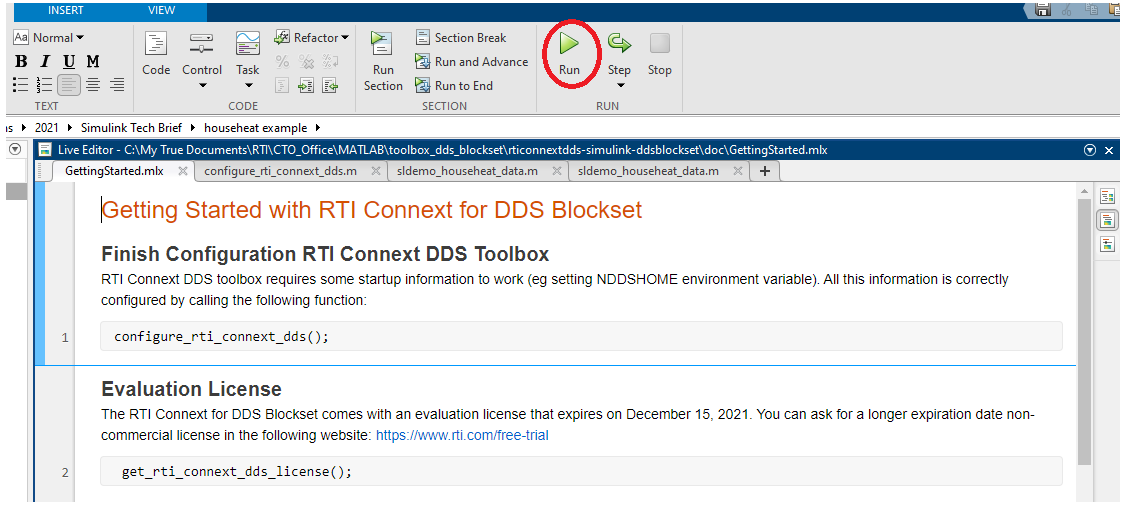

```
// Send: '<Root>/Write DDS Sample'
DataWriterSendData.SendData(&rtb_Output_BusCreator_BusCreato,
```

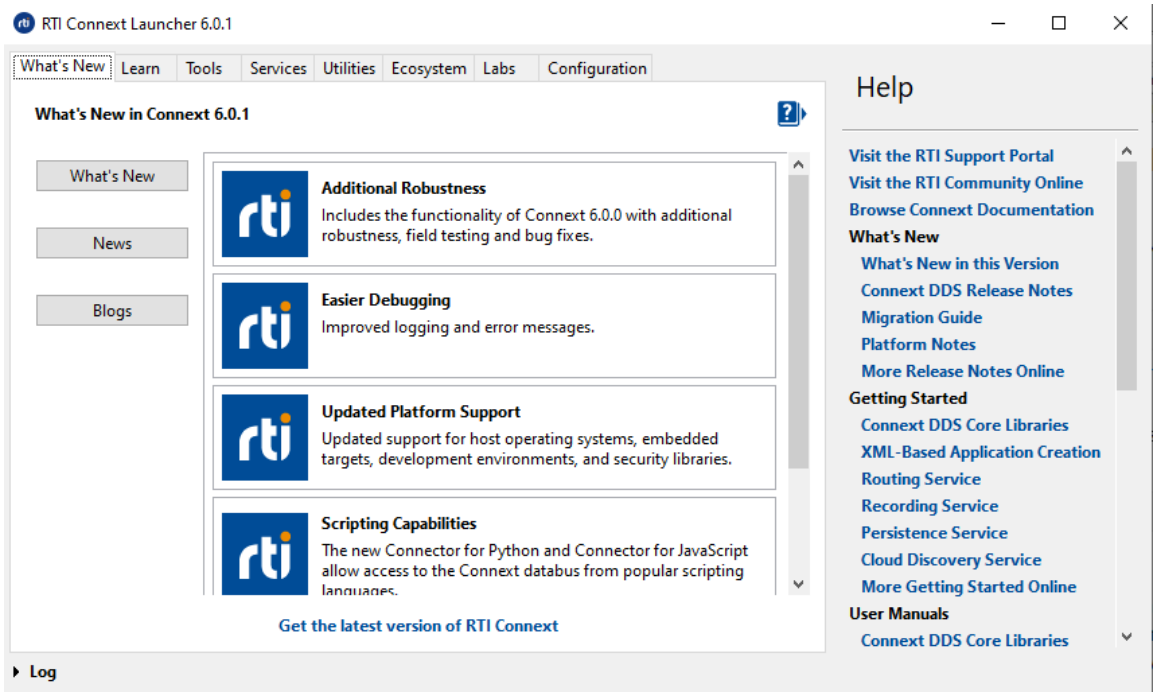


RTI Connext for DDS Blockset

```
run_rti_launcher();
```







- 这是RTI官方提供的MATLAB插件，含2个月试用License
- 即使之前从未安装RTI Connext，该插件也提供了必要的组件，帮助MATLAB用户调试和优化DDS系统
- 对于已经安装RTI Connext Professional的用户，不必再安装该插件，只需按说明配置环境变量，即可开展工作

https://www.mathworks.com/matlabcentral/fileexchange/94665-rti-connext-dds-connectivity-framework-for-dds-blockset

将C++代码集成到SOA框架 – DDS 工作流程

R2021a

